

Practical–2: Study of Different Types of Network Cables

Aim:

To study different types of network cables and understand their construction, characteristics, advantages, disadvantages, and applications. Also, to construct Ethernet straight-through and crossover cables.

1. Types of Network Cables

The major types of network cables are:

1. **Unshielded Twisted Pair (UTP) Cable**
2. **Shielded Twisted Pair (STP) Cable**
3. **Coaxial Cable**
4. **Fibre Optic Cable**

Cable Type	Category/Type	Typical Data Rate	Advantages	Disadvantages	Applications
UTP	Cat 3	Up to 10 Mbps	Low cost, easy to install, flexible	More susceptible to EMI and noise	10BASE-T Ethernet
UTP	Cat 5	Up to 100 Mbps	Low cost, easy installation	Older standard, more interference than shielded cables	Fast Ethernet
UTP	Cat 5e	Up to 1 Gbps	Affordable, widely available	Less protection from EMI than shielded cable	Fast Ethernet, Gigabit Ethernet
STP	Cat 6 / Cat 6a	Up to 10 Gbps*	Shielding reduces noise and interference	More expensive, installation is more difficult	Gigabit Ethernet, 10G Ethernet
SSTP	Cat 7	Up to 10 Gbps	Excellent shielding, high performance	Expensive and comparatively difficult to install	Gigabit Ethernet, 10G Ethernet
Coaxial	RG-59, RG-6, RG-11	Depends on application	Good shielding, reliable,	Bulky, limited flexibility	TV networks, broadband Internet

			relatively low signal loss		
Fibre Optic	Single-mode	Very high speed	High bandwidth, long distance, secure, immune to EMI	Expensive, requires skilled installation	WANs, data centres, backbone networks
Fibre Optic	Multimode	Very high speed	High bandwidth, suitable for shorter distances	More distance-limited than single-mode	LANs, data centres

*The actual supported distance and speed depend on the exact Ethernet standard and cable installation.

2. UTP Cable

UTP (Unshielded Twisted Pair) consists of pairs of insulated copper wires twisted together. The twisting helps reduce electromagnetic interference and crosstalk.

Advantages:

- Low cost
- Easy to install
- Lightweight and flexible
- Small overall diameter

Disadvantages:

- More susceptible to electromagnetic interference than shielded cables
- Limited transmission distance for high-speed Ethernet

Applications:

- Ethernet LANs
- Telephone networks
- Fast Ethernet
- Gigabit Ethernet

3. STP Cable

STP (Shielded Twisted Pair) is similar to UTP but contains additional metallic shielding around the wire pairs or cable.

Advantages:

- Better protection against EMI
- Reduced interference and crosstalk
- Suitable for electrically noisy environments

Disadvantages:

- More expensive than UTP
- More difficult to install
- Requires proper grounding

Applications:

- Gigabit Ethernet
- 10-Gigabit Ethernet
- Industrial networks
- Data centres

4. Coaxial Cable

A **coaxial cable** contains a central conductor, insulating layer, metallic shield, and outer protective jacket.

Common types include:

- **RG-59**
- **RG-6**
- **RG-11**

Advantages:

- Good resistance to interference
- Reliable signal transmission
- Suitable for broadband applications

Disadvantages:

- Bulky compared with twisted-pair cable
- Less flexible
- Installation can be more difficult

Applications:

- Television networks
- Cable Internet
- CCTV systems
- RF communication

5. Fibre Optic Cable

A **fibre optic cable** transmits information using pulses of light through glass or plastic optical fibre rather than electrical signals.

The two major types are:

1. **Single-mode fibre**
2. **Multimode fibre**

Advantages:

- Very high bandwidth
- High transmission speed
- Long-distance communication
- Immune to electromagnetic interference
- High security

Disadvantages:

- Higher installation cost
- Requires specialised tools and skilled technicians
- Fibre can be more delicate than copper cable

Applications:

- Internet backbone

- Data centres
- Long-distance communication
- Telecommunications
- High-speed networks

b) Making an Ethernet Straight-Through / Crossover Cable

Tools and Materials Required

- CAT5/CAT5e Ethernet cable
- RJ45 connectors – 2
- Crimping tool
- Cable stripper/cutter
- Cable tester
- RJ45 connector boots/shields, if required

Difference Between Straight-Through and Crossover Cable

Straight-through cable:

Both ends use the **same wiring standard**.

Example:

End A – T568A

End B – T568A

or

End A – T568B

End B – T568B

Crossover cable:

The two ends use **different wiring standards**.

End A – T568A

End B – T568B

T568A Colour Sequence

1. White-Green
2. Green
3. White-Orange
4. Blue
5. White-Blue
6. Orange
7. White-Brown
8. Brown

T568B Colour Sequence

1. White-Orange
2. Orange
3. White-Green
4. Blue
5. White-Blue
6. Green
7. White-Brown
8. Brown

Procedure

Step 1:

Thread the protective boot/shield onto the Ethernet cable if one is being used.

Step 2:

Using the stripping section of the crimping tool, carefully remove approximately **1.5 cm** of the outer cable jacket.

Step 3:

Separate and untwist the four twisted pairs. Arrange the eight individual wires according to either the **T568A** or **T568B** colour sequence.

Step 4:

Straighten the wires and place them side by side in the correct order. Trim the wire ends evenly.

Step 5:

Hold the RJ45 connector with the contacts facing upward and carefully insert all eight wires into the connector. Make sure the wires remain in the correct order.

Step 6:

Insert the RJ45 connector into the crimping tool and press firmly. The crimping action secures the wires to the connector contacts.

Step 7:

Prepare the other end of the cable.

- For a **straight-through cable**, use the **same standard on both ends**.
- For a **crossover cable**, use **T568A on one end and T568B on the other end**.

Step 8:

Use an Ethernet cable tester to verify the cable. Check that all eight conductors are correctly connected.

Result

The different types of network cables—**UTP, STP, coaxial, and fibre optic cables**—were studied. An Ethernet **straight-through/crossover cable** was constructed using an RJ45 connector and crimping tool, and its connectivity was tested successfully.